

of ozone depletion may be severe for plants, animals, and microorganisms. Some scientists expect increases in skin cancer and cataracts in humans, as well as unpredictable effects on crops and natural communities, especially the phytoplankton that are responsible for much of Earth's primary production. Such effects have been documented in field experiments. For example, scientists have shown that the growth of phytoplankton decreases and damage to their DNA increases when the ozone hole opens over the Southern Ocean (Antarctic Ocean) each year.

The good news about the ozone hole is how quickly many countries have responded to it. Since 1987, at least 197 nations, including the United States, have signed the Montreal Protocol, a treaty that regulates the use of ozone-depleting chemicals. Most nations have ended the production of CFCs. As a consequence, chlorine concentrations in the stratosphere have stabilized and ozone depletion is slowing. But even though CFC emissions today are close to zero, chlorine molecules already in the atmosphere will continue to influence stratospheric ozone levels for at least 50 years.

The partial destruction of Earth's ozone shield is one more example of how greatly humans can disrupt the dynamics of ecosystems and the biosphere. It also highlights our ability to solve environmental problems when we set our minds to it. However, in 2018 researchers detected an unexpected rise in atmospheric CFC levels, suggesting that unreported new production of these compounds may be occurring. Thus, once environmental regulations are in place, we cannot take them for granted—ongoing monitoring and enforcement are needed.

CONCEPT CHECK 56.4

1. How can the addition of excess mineral nutrients to a lake threaten its fish population?
2. **MAKE CONNECTIONS** There are vast stores of carbon-containing organic matter in the soils of northern coniferous forests and tundra around the world. Suggest why scientists who study global warming are closely monitoring these carbon reservoirs (see Figure 55.14).
3. **MAKE CONNECTIONS** Mutagens are chemical and physical agents that induce mutations in DNA (see Concept 17.5). How does reduced ozone concentration in the atmosphere increase the likelihood of mutations in various organisms?

For suggested answers, see Appendix A.

CONCEPT 56.5

Sustainable development can improve human lives while conserving biodiversity

With the increasing loss and fragmentation of habitats, changes in Earth's physical environment and climate, and increasing human population (see Concept 53.6), we face difficult trade-offs in managing the world's resources. Preserving all habitat patches isn't feasible, so biologists must help societies

set conservation priorities by identifying which habitat patches are most crucial, while likewise improving the quality of life for local people. Ecologists use the concept of *sustainability* as a tool to establish long-term conservation priorities.

Sustainable Development

We need to understand the interconnections of the biosphere to protect species from extinction and improve the quality of human life. To this end, many nations, scientific societies, and other groups have embraced the concept of **sustainable development**, economic development that meets the needs of people today without limiting the ability of future generations to meet their needs.

Achieving sustainable development is an ambitious goal. To sustain ecosystem processes and stem the loss of biodiversity, we must connect life science with the social sciences, economics, and the humanities. We must also reassess our personal values. Those of us living in wealthier nations have a larger ecological footprint than do people living in developing nations (see Concept 53.6). By including the long-term costs of consumption in our decision-making processes, we can learn to value the ecosystem services that sustain us.

Case Study: Sustainable Development in Costa Rica

The success of conservation in Costa Rica discussed earlier has required a partnership between the government, nongovernment organizations (NGOs), and private citizens. Many nature reserves established by individuals have been recognized as national wildlife reserves and given significant tax benefits. However, conservation and restoration are only one facet of sustainable development; the other is improving the human condition.

How have the living conditions of the Costa Rican people changed as the country has pursued its conservation goals? As represented in **Figure 56.35**, two fundamental indicators

▼ **Figure 56.35** Infant mortality and life expectancy at birth in Costa Rica.

